Program / Source Code

```
#include <stdio.h>
#include "billing.h"

long long calculateBill(int units)
{
   if(units <= FREE_UNITS)
      return 0;
  if(units <= 200)
    return (units - 100) * RATE1;

   return 100 * RATE1 + (units - 200) * RATE2;
}

int main()
```

```
{
   int units = 250;

   printf("Units = %d\n", units);
   printf("Bill = Rs. %.2f\n",
        calculateBill(units) / 100.0);

#ifdef DEBUG
   printf("Debug mode is ON\n");
#endif
   return 0;
}
```

Output Screenshot



Program / Source Code

```
#include <stdio.h>

int main()
{
    int household, previous, current, units;
    long long bill;

    printf("Enter household number: ");
    if(scanf("%d", &household) != 1)
        return 1;

    printf("Enter previous reading: ");
    if(scanf("%d", &previous) != 1)
        return 1;

    printf("Enter current reading: ");
    if(scanf("%d", &current) != 1)
        return 1;

    if(previous < 0 || previous > 65535 ||
       current < 0 || current > 65535)
    {
        printf("Invalid reading");
        return 1;
    }
```

```
    if(current >= previous)
        units = current - previous;
    else
        units = (65535 - previous) + 1 + current;

    if(units <= 100)
        bill = 0;
    else if(units <= 200)
        bill = (units - 100) * 225LL;
    else
        bill = 100 * 225LL + (units - 200) * 450LL;

    printf("\nHousehold Number: %d", household);
    printf("\nUnits Consumed: %d", units);
    printf("\nEnergy Charge: Rs. %lld.%02lld\n",
           bill / 100, bill % 100);

    return 0;
}
```

Output

```
Enter household number: 101
Enter previous reading: 500
Enter current reading: 650
Enter category: A

Household Number: 101
Previous Reading: 500
Current Reading: 650
Units Consumed: 150
Category: A
```

Program / Source Code

```
#include <stdio.h>

int main()
{
    int units;
    long long charge;

    printf("Enter units: ");
    scanf("%d", &units);

    if(units <= 100)
        charge = 0;
    else if(units <= 200)
        charge = (long long)(units - 100) * 225;
    else
        charge = 100LL * 225 +
                 (long long)(units - 200) * 450;

    printf("Energy Charge: Rs. %lld.%02lld\n",
           charge / 100, charge % 100);
```

Output

```
Enter units: 250
Energy Charge: Rs. 450.00
```

Program / Source Code:

```
#include <stdio.h>

int main()
{
   int units = 250;
   int charge;

   charge = 100 * 225 + (units - 200) * 450;

   printf("Units = %d\n", units);
   printf("Charge = %d paise\n", charge);
```

Output




```
Units = 250
Charge = 45000 paise
```